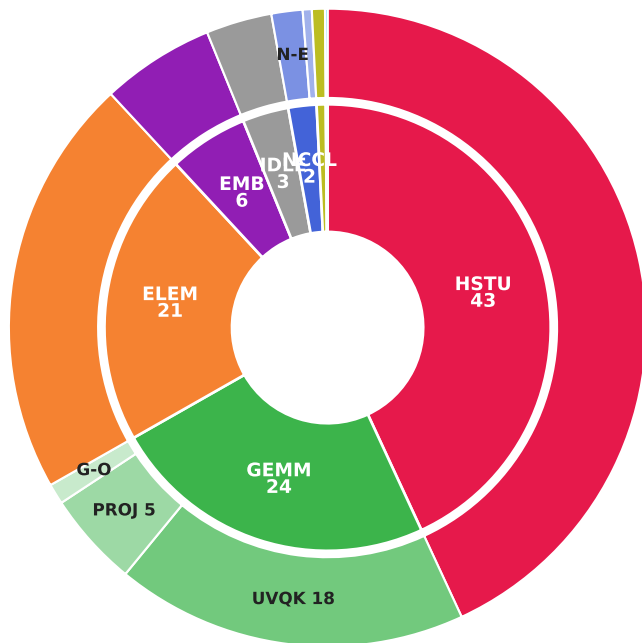
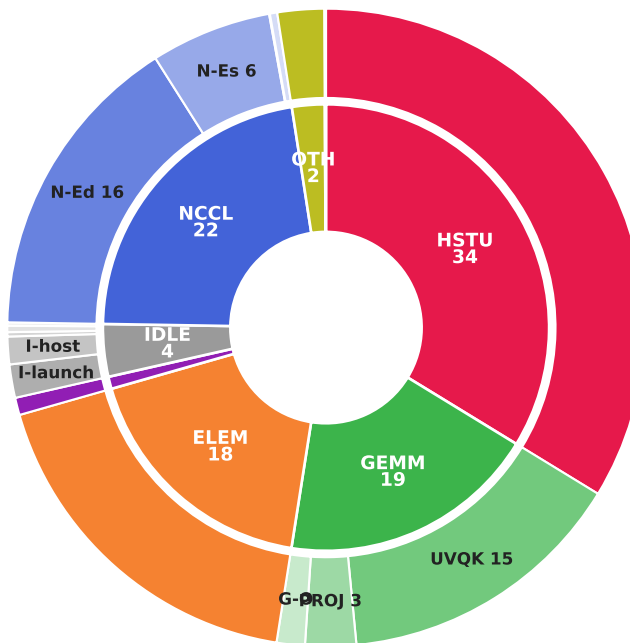


HSTU GPU-time sunburst — NON-JAGGED (matched config), exposed accounting, fastest step (upstream \$2.2 method)
 inner ring = grouping · outer ring = leaves · % of the step

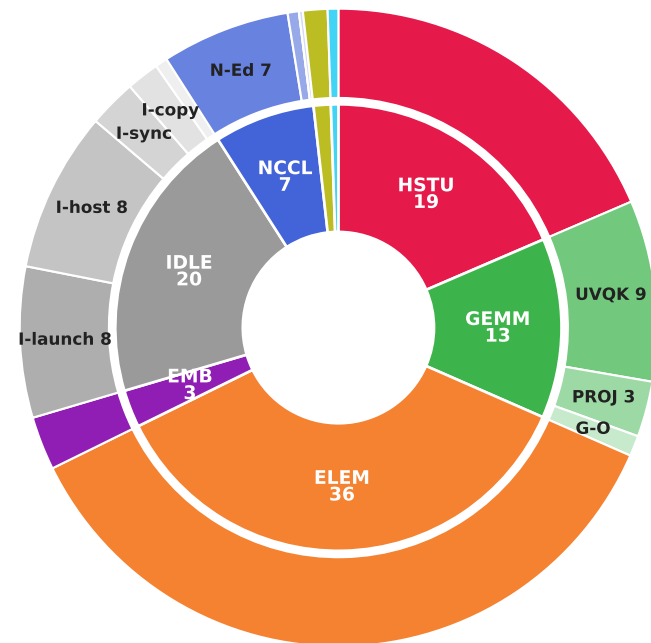
Upstream H100 · step 162 · D256 (nj)



Our H100 · step 159 · 235.6 ms · D256 (nj)



Our GB300 · step 153 · 78.0 ms · D128 (nj)



- | | | | | | |
|----------------------------|-------------------------------------|---|---|----------------------------------|--------------------|
| ■ HSTU = hstu fwd/bwd | ■ G-O = gemm / others | ■ EMB = embedding op | ■ N-Os = nccl sparse (overlap) | ■ I-host = idle: host/python gap | ■ OTH = others |
| ■ UVQK = gemm / uvqk | ■ GEMM = gemm (ours: not sub-split) | ■ N-Ed = nccl DENSE/all-reduce (exposed) | ■ N-E = nccl exposed (upstream, unsplit) | ■ I-sync = idle: sync wait | ■ OVL = overlapped |
| ■ PROJ = gemm / projection | ■ ELEM = elementwise | ■ N-Es = nccl SPARSE/all-to-all (exposed) | ■ I-launch = idle: kernel-launch (CPU dispatch) | ■ I-copy = idle: H↔D copy | |